

ST CYPRIAN HIGH SCHOOL KYABAKADDE.

S.2 MATHEMATICS

HOLIDAY PACKAGE.

ITEM 1

Simon is the district Inspector of schools at the district. He owns a farm that measures 308ft by 228ft. He plans to put fencing posts along its sides at equal greatest distances apart. He needs to know the number of poles he needs to achieve this.

One day, Simon discovered that the casual laborer's he hired had used a third of the farm for banana growing, one quarter for coffee growing and two thirds of the remainder were used for mixed farming. He wants to double his investment on the farming that carries the biggest percentage.

On a particular day, Simon invited 1200 students on his farm for a field study. 55% of the students who attended the study were boys. In his interview, Simon told students that the price of his farm was valued at forty million one hundred twenty-five thousand shillings. It increased by 25% after the first year and by 10% in the second year.

Task

- (a) Help Simon know the number of fencing posts he needs.
- (b) Help Simon to know the type of farming whose investment he will double.
- (c) Find the number of girls that attended the field study on his farm.
- (d) The value of Simon's farm at the end of the second year.

ITEM 2

A certain teacher has 40 girls and 30 boys in his class. He wants to group them in such a way that each group has an equal number of boys and equal number of girls. So, he requests you to advise him on the highest number of groups he can form including the boys and girls that can be in each of the groups.

He gave an assignment that was marked out of 40. For one of the students who scored 25, the parent is asking what the learner scored out of 100 since that is the language he understands.

The teacher plans to take students for a tour. 30 students paid fifty thousand Ugandan shillings each. Two fifth of the total amount of money collected is planned for transport for all children. The transport company he contacted asked for Ugx.600, 000 for the whole journey

they plan to travel. Now he is wondering if they can afford it.

Task:

- a) Help the teacher determine the number of groups and the number of girls and boys that will be in each group.
- b) Help the teacher to reply the parent of the student.
- c) Will he afford the transport company he contacted?

ITEM 3

Garden City Shopping Mall in Kampala is experiencing a massive turnout of customers, leading to a critical shortage of parking spaces. To address this challenge, the mall's management has planned an innovative solution: - a triangular rooftop parking design, exclusively designed for compact cars, each spanning an area of 0.2 square units, to cater for the growing demand of space. The parking area is bounded by the following constraints; $x + y < 3$, $x - y + 3 \geq 0$, and $y + 1 \geq 0$. To capitalize on the high demand for parking, the mall plans to impose a parking fee of Shs 2500 per vehicle, with calculations indicating that on peak days, the cars will occupy a total area of 12.4 square units, yielding a significant income source.

TASK

- a) (i) Create a graphical representation on paper to help Garden City management accurately visualize and understand the layout and boundaries of the new parking design.
- (ii) Assist the management in identifying the accurate coordinates and size of the new parking area in square units.
- (ii) Will the new design accommodate the expected maximum number of vehicles? If so, determine the maximum number of cars that can be accommodated in the new parking area at full capacity.
- b) Forecast the mall's daily highest revenue on a peak day.

ITEM 4

Molly started a wholesale business while still at her parent's home. Below is how she plans to use her net profits.

- 20% of the net profits to be re-invested in the business.

- Part of the balance is to be saved in her savings account and the rest is to be for her major personal expenses. This will be in the ratio of 1:3 respectively.

She plans to move out of her parent's house if the portion of her net profits that she plans to spend on her major personal expenses can cover all of them. Her major expected personal expenses are; house rent of about UGX.300,000, and groceries of about UGX. 200,000 and transport of about UGX. 100,000. The business made a net profit of one million five hundred thousand Uganda shillings in the first month.

She bought 36 bars of bathing soap and 18 packets of detergent that were all to be packed by her worker in such a way that each package contained the same number of both items without a remainder. The packages were to be given to those who bought goods in plenty as gifts. All packages were given out before Molly got to know the highest number of packages that her worker was able to make out of the bought items.

TASK:

- a) (i) How much of her net profit does she plan to re-invest in the business?
- b) (ii) How much of her net profit does she plan to spend on personal expenses?
(iii) Will Molly move out of her parents' house? Justify your answer.
- c) Help Molly to know the highest number of packages her worker was able to make out of the items she bought.

ITEM 5

Your father is one of the organizers of a marathon they want to draw the map of the route that the participants will during the race.

At their chosen starting point they chose to take a road that turned $E30^{\circ}S$ and they moved for 5km where they set up point **B** which will be used as a checkpoint, they then turned through 235° moving a distance of 9km to point **C** which will be the finishing point, however on returning to the office they decided that the finishing point should be put at point **A** to cut costs of organizing the two places but they were not sure of the details of that route from **C** to **A** that had to be included on the map.

They want to hire a vehicle that will be used to film the racers, the vehicle available consumes 2 litres per km and a litre costs UGX.5500, the owner of the vehicle has asked for UGX.500,000 plus fueling the vehicle for the total distance to be covered but the vehicle owner plans to buy fuel from the fuel station where he is given a discount of 5% for every 100,000 worth of fuel he buys since he is a regular customer.

TASK;

- (a) Help your father determine the direction from point C to the new finishing point A that will be shown on the map to be drawn.
- (b) (i) You are required to determine the total cost of hiring the vehicle.
(ii) Do you think the vehicle owner will save some money on fuel if so how much?

OTHERS

1. The difference between the squares of two numbers is 40. If the larger number exceeds the smaller by 4, find the two numbers.

[Ans: 7 and 3]

2. A man is now four times as old as his son. Eight years ago, the product of their ages was 220 years. Find the son's present age.

[Ans: 13 years]

3. Jane is three years older than Peace. In two years time, the product of their ages will be 238. How old is Jane now?

[Ans: 15 years]

END

FOR INQUIRIES.

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